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Cut rope to maximise product

Difficulty: **Medium**Accuracy: **40.16%**Submissions: **17K+**Points: **4**

Given a rope of length **n** meters, cut it into multiple smaller ropes such that the product of their lengths is maximized. At least one cut is mandatory.

Examples:

Input: n = 2**Output:** 1**Explanation:** Since 1 cut is mandatory. Maximum obtainable product is **1 * 1 = 1**.**Input:** n = 5**Output:** 6**Explanation:** Maximum obtainable product is **2 * 3 = 6**.

Constraints:

 $2 \leq n \leq 58$ [Try more examples](#)

```
1 class Solution:
2     def maxProduct(self, n):
3         if n == 2:
4             return 1
5         if n == 3:
6             return 2
7
8         dp = [0] * (n + 1)
9         dp[1] = 1
10        dp[2] = 2
11        dp[3] = 3
12
13        for i in range(4, n + 1):
14            max_prod = 0
15            for j in range(1, i // 2 + 1):
16                max_prod = max(max_prod, dp[j] * dp[i - j])
17            dp[i] = max_prod
18
19        return dp[n]
```

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